Tutorial

DATE 19 | 1 | 26

Q. Write a Regular expression to accept a language L over a, b.

1. Which is starting with a

→ $a(a+b)^*$

2. Ending with a

→ $(a+b)^* a$

3. Starting with a & ending with b

→ $a(a+b)^* b$

4. Contains a substring abb

→ $(a+b)^*$ abb $(a+b)^*$

Q. Construct RE for a language L over 0, 1 in which every 0 is immediately followed atleast two 1's.

→ $1^* (011)^* 1^*$

Q. Construct a RE which

1. Contains the string having a number of 0's are multiple of 3.

→ $1^* (01^* 01^* 01^*)^*$

2. Length of string is even.

→ $((0+1)(0+1))^*$

3. Length of string should be odd.

→ $(0+1)((0+1)(0+1))^*$

4. Number of A's should be even or multiple of 2.

→ $(b^* ab^* a)^* b^*$

5. Number of 0 are even and number of 1 even.

→ ~~(00+~~ $(00|11|(01|10)(00|11)^*(01|10))^*$

6. Number of 0's odd and number of 1's odd.

→ $0(00|11|(01|10)(00|11)^*(01|10))^*1|1(00|11|(01|10)(00|11)^*(01|10))^*0$

7. Number of 0's odd and number of 1's even.

→ $(00|11|(01|10)(00|11)^*(01|10))^*(0|1(00|11)^*(01|10))$

8. Number of 0's even and number of 1's odd.

→ $(00|11|(01|10)(00|11)^*(01|10))^*(1|0(00|11)^*(01|10))$

12/2026

Q. Construct the context free grammar for a language L over a, b where $L_1 = \{a^n b^n \mid n \geq 1\}$

$L_2 = \{a^n b^{2n} \mid n \geq 1\}$

$L_3 = \{ww^T \mid w \in \{a,b\}^*\}$

$L_4 = \{wcw^T \mid w \in \{a,b\}^*\}$

$L_5 = \{a^n b^n c^j \mid n \geq 1, j \geq 0\}$

$L_6 = \{c^e a^n b^n \mid n \geq 1, e \geq 0\}$

① $L_1 = \{a^n b^n \mid n \geq 1\}$

$S \rightarrow ab \mid aSb$

$G_1 = \{S\}, \{a,b\}, P, \{S\}$

$\{S\}$	$\{a,b\}$	P	$\{S\}$
variable	input	production rule	start.

② $L_2 = \{a^n b^{2n} \mid n \geq 1\}$

$S \rightarrow abb \mid aSbb$

$G_2 = \{S\}, \{a, b\}, P, \{S\}$

③ $L_3 = \{ WW^T \mid W \in \{a, b\}^* \} \rightarrow$ (even palindrome string)

~~$S \rightarrow$~~ abba, aabbba, abbbba

$S \rightarrow aSb \mid bSb \mid \epsilon$

$G_3 = \{S\}, \{\epsilon, a, b\}, P, \{S\}$

④ $L_4 = \{ WCW^T \mid W \in \{a, b\}^* \} \rightarrow$ (odd palindrome string)

$S \rightarrow aSa \mid bSb \mid c$

⑤ $L_5 = \{ a^n b^n c^j \mid n \geq 1, j \geq 0 \}$

ab, abc, aabc,

$S \rightarrow AC$

$A \rightarrow aAb \mid ab$

$C \rightarrow cC \rightarrow \epsilon$

$S \rightarrow ab \mid aSb \mid Sc$ ← if c is present it should be last

⑥ $L_6 = \{ c^e a^n b^n \mid n \geq 1, e \geq 0 \}$

~~ab~~, ~~aab~~, ab, cab, aabb

$S \rightarrow ab \mid aSb \mid cS$ ← c should be at starting.

⑦ $L_7 = \{ a^n b^n \mid n \geq 0 \}$

$S \rightarrow \epsilon \mid aSb$

$G_7 = \{S\}, \{a, b\}, P, \{S\}$